

Production service portal

SYNTHETIC — DEMONSTRATION ONLY

Accepted semantic revision: 0

Design at a glance

This design contains 7 components and 4 connections. The sections below describe the accepted model; unspecified facts stay undecided.

Design notes

Synthetic authored example. Confirmed claims mean confirmed within this fictional reference; they are not organisational approval. See Sources for the complete specification and open decisions.

What runs where

- Employee workstation [employee] is a employee access in Client · user devices. Status: proposed; 1 declared instance. Form: physical.
- Service desk workstation [service-desk] is a service desk access in Client · user devices. Status: proposed; 1 declared instance. Form: physical.
- Portal web service [portal-web] is a web frontend in Z1 · application services. Status: proposed; 1 declared instance. Form: virtual. Runs on Application hosts.
- Request application [portal-app] is a request processing in Z1 · application services. Status: proposed; 2 declared instances. Redundancy: active active. Form: virtual. Runs on Application hosts.
- Application hosts [app-host] is a application compute host in Z1 · application services. Status: proposed; 2 declared instances. Redundancy: active active. Form: physical.
- Zone firewall [zone-firewall] is a zone enforcement in Z1 · application services. Status: proposed; 1 declared instance. Form: virtual. Runs on Application hosts.
- Request database [request-db] is a protected relational data and audit records in Z2 · protected data. Status: proposed; 2 declared instances. Redundancy: active passive. Form: physical.

Connections

- Employee workstation → Portal web service [employee-web]: service requests, using HTTPS (port 443); Employee workstation starts the session. Data flow: bidirectional. Enforcement: not recorded.
- Service desk workstation → Portal web service [desk-web]: service desk actions, using HTTPS (port 443); Service desk workstation starts the session. Data flow: bidirectional. Enforcement: not recorded.
- Portal web service → Request application [web-app]: request processing, using HTTPS (port 443); Portal web service starts the session. Data flow: bidirectional. Enforcement: not recorded.
- Request application → Request database [app-database]: database access, using PostgreSQL (port 5432); Request application starts the session. Data flow: bidirectional. Enforcement: Zone firewall.

Requirements and open decisions

- no internet: True.
- resilience required: True.
- availability strategy: undecided.
- equipment sizing: undecided.
- client direct database access: forbidden.
- What end-to-end availability strategy should be used for the portal? Not decided.

Sources and review trail

- [claim-portal-users] confirmed — techspec.docx, version 1.0, paragraph/6: [P01] The Production service portal has three trust tiers: Client · user devices; Z1 · application services; and Z2 · protected data. Every item below is a proposed part of this fictional design. Zone placement describes intended trust separation, not proof of firewall configuration. The model groups responsibility as User access, Portal services and Protected data.
- [claim-portal-services] confirmed — techspec.docx, version 1.0, paragraph/6: [P01] The Production service portal has three trust tiers: Client · user devices; Z1 · application services; and Z2 · protected data. Every item below is a proposed part of this fictional design. Zone placement describes intended trust separation, not proof of firewall configuration. The model groups responsibility as User access, Portal services and Protected data.
- [claim-portal-data] confirmed — techspec.docx, version 1.0, paragraph/6: [P01] The Production service portal has three trust tiers: Client · user devices; Z1 · application services; and Z2 · protected data. Every item below is a proposed part of this fictional design. Zone placement describes intended trust separation, not proof of firewall configuration. The model groups responsibility as User access, Portal services and Protected data.
- [claim-client] confirmed — techspec.docx, version 1.0, paragraph/6: [P01] The Production service portal has three trust tiers: Client · user devices; Z1 · application services; and Z2 · protected data. Every item below is a proposed part of this fictional design. Zone placement describes intended trust separation, not proof of firewall configuration. The model groups responsibility as User access, Portal services and Protected data.
- [claim-z1] confirmed — techspec.docx, version 1.0, paragraph/6: [P01] The Production service portal has three trust tiers: Client · user devices; Z1 · application services; and Z2 · protected data. Every item below is a proposed part of this fictional design. Zone placement describes intended trust separation, not proof of firewall configuration. The model groups responsibility as User access, Portal services and Protected data.
- [claim-z2] confirmed — techspec.docx, version 1.0, paragraph/6: [P01] The Production service portal has three trust tiers: Client · user devices; Z1 · application services; and Z2 · protected data. Every item below is a proposed part of this fictional design. Zone placement describes intended trust separation, not proof of firewall configuration. The model groups responsibility as User access, Portal services and Protected data.
- [claim-employee] confirmed — techspec.docx, version 1.0, paragraph/7: [P02] Client contains one physical Employee workstation and one physical Service desk workstation. They are user devices only. No application server or database is placed in Client.

- [claim-service-desk] confirmed — techspec.docx, version 1.0, paragraph/7: [P02] Client contains one physical Employee workstation and one physical Service desk workstation. They are user devices only. No application server or database is placed in Client.
- [claim-portal-web] confirmed — techspec.docx, version 1.0, paragraph/8: [P03] Z1 contains one virtual Portal web service and two virtual Request application instances operating active-active. Both services run on the two physical Application hosts in Z1. The host pair is specified as active-active; its hardware capacity is not yet sized. One virtual Zone firewall also runs on an Application host; firewall failover is undecided.
- [claim-portal-app] confirmed — techspec.docx, version 1.0, paragraph/8: [P03] Z1 contains one virtual Portal web service and two virtual Request application instances operating active-active. Both services run on the two physical Application hosts in Z1. The host pair is specified as active-active; its hardware capacity is not yet sized. One virtual Zone firewall also runs on an Application host; firewall failover is undecided.
- [claim-app-host] confirmed — techspec.docx, version 1.0, paragraph/8: [P03] Z1 contains one virtual Portal web service and two virtual Request application instances operating active-active. Both services run on the two physical Application hosts in Z1. The host pair is specified as active-active; its hardware capacity is not yet sized. One virtual Zone firewall also runs on an Application host; firewall failover is undecided.
- [claim-zone-firewall] confirmed — techspec.docx, version 1.0, paragraph/8: [P03] Z1 contains one virtual Portal web service and two virtual Request application instances operating active-active. Both services run on the two physical Application hosts in Z1. The host pair is specified as active-active; its hardware capacity is not yet sized. One virtual Zone firewall also runs on an Application host; firewall failover is undecided.
- [claim-request-db] confirmed — techspec.docx, version 1.0, paragraph/9: [P04] Z2 contains two physical Request database servers operating active-passive. The database stores requests and application audit records. No Client device may connect directly to the database.
- [claim-deployment-employee] confirmed — techspec.docx, version 1.0, paragraph/7: [P02] Client contains one physical Employee workstation and one physical Service desk workstation. They are user devices only. No application server or database is placed in Client.
- [claim-deployment-service-desk] confirmed — techspec.docx, version 1.0, paragraph/7: [P02] Client contains one physical Employee workstation and one physical Service desk workstation. They are user devices only. No application server or database is placed in Client.
- [claim-deployment-portal-web] confirmed — techspec.docx, version 1.0, paragraph/8: [P03] Z1 contains one virtual Portal web service and two virtual Request application instances operating active-active. Both services run on the two physical Application hosts in Z1. The host pair is specified as active-active; its hardware capacity is not yet sized. One virtual Zone firewall also runs on an Application host; firewall failover is undecided.
- [claim-deployment-portal-app] confirmed — techspec.docx, version 1.0, paragraph/8: [P03] Z1 contains one virtual Portal web service and two virtual Request application instances operating active-active. Both services run on the two physical Application hosts in Z1. The host pair is specified as active-active; its hardware capacity is not yet sized. One virtual Zone firewall also runs on an Application host; firewall failover is undecided.

- [claim-deployment-app-host] confirmed — techspec.docx, version 1.0, paragraph/8: [P03] Z1 contains one virtual Portal web service and two virtual Request application instances operating active-active. Both services run on the two physical Application hosts in Z1. The host pair is specified as active-active; its hardware capacity is not yet sized. One virtual Zone firewall also runs on an Application host; firewall failover is undecided.
- [claim-deployment-zone-firewall] confirmed — techspec.docx, version 1.0, paragraph/8: [P03] Z1 contains one virtual Portal web service and two virtual Request application instances operating active-active. Both services run on the two physical Application hosts in Z1. The host pair is specified as active-active; its hardware capacity is not yet sized. One virtual Zone firewall also runs on an Application host; firewall failover is undecided.
- [claim-deployment-request-db] confirmed — techspec.docx, version 1.0, paragraph/9: [P04] Z2 contains two physical Request database servers operating active-passive. The database stores requests and application audit records. No Client device may connect directly to the database.
- [claim-employee-web] confirmed — techspec.docx, version 1.0, paragraph/12: [P05] Each workstation initiates an HTTPS session to Portal web service on TCP 443 for service requests or service-desk actions. These are application sessions, not infrastructure administration sessions.
- [claim-desk-web] confirmed — techspec.docx, version 1.0, paragraph/12: [P05] Each workstation initiates an HTTPS session to Portal web service on TCP 443 for service requests or service-desk actions. These are application sessions, not infrastructure administration sessions.
- [claim-web-app] confirmed — techspec.docx, version 1.0, paragraph/13: [P06] Portal web service initiates HTTPS sessions to Request application on TCP 443. Request application initiates PostgreSQL sessions to Request database on TCP 5432. Zone firewall is the explicit enforcement reference for the Z1-to-Z2 database interface; its intended rule permits only the application service identity to that database service and denies other cross-tier traffic.
- [claim-app-database] confirmed — techspec.docx, version 1.0, paragraph/13: [P06] Portal web service initiates HTTPS sessions to Request application on TCP 443. Request application initiates PostgreSQL sessions to Request database on TCP 5432. Zone firewall is the explicit enforcement reference for the Z1-to-Z2 database interface; its intended rule permits only the application service identity to that database service and denies other cross-tier traffic.
- [claim-portal-offline] confirmed — techspec.docx, version 1.0, paragraph/15: [P07] The application is internal and has no required internet-hosted dependency. Data remains on premises. Application database and audit records are the only stores defined here; backups, audit retention, IP addresses, VLAN numbers, hardware capacity and infrastructure administration remain undecided.
- [claim-portal-resilience] confirmed — techspec.docx, version 1.0, paragraph/16: [P08] A service availability requirement exists, but its end-to-end strategy is not yet approved: the application and database have the stated redundancy, while the web service and virtual firewall remain single instances. The service owner must choose a recovery target and resolve these remaining failure points before production acceptance.
- [claim-portal-availability] confirmed — techspec.docx, version 1.0, paragraph/16: [P08] A service availability requirement exists, but its end-to-end strategy is not yet approved: the application and database have the stated redundancy, while the web service and virtual firewall remain single instances. The service owner must choose a recovery target and resolve these remaining failure points before production acceptance.

- [claim-portal-sizing] confirmed — techspec.docx, version 1.0, paragraph/15: [P07] The application is internal and has no required internet-hosted dependency. Data remains on premises. Application database and audit records are the only stores defined here; backups, audit retention, IP addresses, VLAN numbers, hardware capacity and infrastructure administration remain undecided.
- [claim-portal-boundary] confirmed — techspec.docx, version 1.0, paragraph/9: [P04] Z2 contains two physical Request database servers operating active-passive. The database stores requests and application audit records. No Client device may connect directly to the database.
- [claim-portal-availability-question] confirmed — techspec.docx, version 1.0, paragraph/16: [P08] A service availability requirement exists, but its end-to-end strategy is not yet approved: the application and database have the stated redundancy, while the web service and virtual firewall remain single instances. The service owner must choose a recovery target and resolve these remaining failure points before production acceptance.

Policy checks and synthetic examples do not constitute organisational approval.